

Supplementary Material

PROOF OF PROPOSITION 1

This section presents the detailed proof of Proposition 1. Let the routing prior $\pi_\phi(\cdot | s_t)$ satisfy $\pi_\phi(k | s_t) > 0$, the variational energy $\bar{E}_k(s_t)$ be composed of reconstruction energy and semantic uncertainty terms, $\tau > 0$, and $q \in \Delta^{K-1}$. The following derivation establishes that the optimal responsibility allocation admits the equivalent softmax expression:

$$q^* = \text{softmax}(\log \pi_\phi(\cdot | s_t) - \bar{\mathbf{E}}(s_t)/\tau). \quad (1)$$

To simplify notation, fix any diffusion state s_t and let

$$\pi_k = \pi_\phi(k | s_t), \quad E_k = \bar{E}_k(s_t). \quad (2)$$

Define the normalizing constant

$$Z = \sum_{j=1}^K \pi_j \exp(-E_j/\tau), \quad (3)$$

and

$$q_k^* = \frac{\pi_k \exp(-E_k/\tau)}{Z}. \quad (4)$$

Since $\pi_k > 0$ and $\exp(-E_k/\tau) > 0$, we have $q_k^* > 0$; moreover $\sum_k q_k^* = 1$, so $q^* \in \Delta^{K-1}$. The above expression is equivalently written as a softmax:

$$q_k^* = \frac{\exp(\log \pi_k - E_k/\tau)}{\sum_j \exp(\log \pi_j - E_j/\tau)}. \quad (5)$$

Consider the variational objective for any $q \in \Delta^{K-1}$:

$$J(q) = \sum_{k=1}^K q_k E_k + \tau \text{KL}(q \| \pi). \quad (6)$$

Expanding the KL term:

$$\begin{aligned} J(q) &= \sum_k q_k E_k + \tau \sum_k q_k \log \frac{q_k}{\pi_k} \\ &= \tau \sum_k q_k \left(\frac{E_k}{\tau} + \log q_k - \log \pi_k \right) \\ &= \tau \sum_k q_k \log \frac{q_k}{\pi_k \exp(-E_k/\tau)}. \end{aligned} \quad (7)$$

From $q_k^* = \pi_k \exp(-E_k/\tau)/Z$, we have $\pi_k \exp(-E_k/\tau) = Z q_k^*$. Substituting:

$$\begin{aligned} J(q) &= \tau \sum_k q_k \log \frac{q_k}{Z q_k^*} \\ &= \tau \text{KL}(q \| q^*) - \tau \log Z, \end{aligned} \quad (8)$$

using $\sum_k q_k = 1$. Since $\text{KL}(q \| q^*) \geq 0$ with equality if and only if $q = q^*$, the global minimum of $J(q)$ over Δ^{K-1} is achieved at $q = q^*$:

$$\min_{q \in \Delta^{K-1}} J(q) = -\tau \log Z. \quad (9)$$

Substituting the definition of Z :

$$-\tau \log Z = -\tau \log \sum_{k=1}^K \pi_\phi(k | s_t) \exp(-\bar{E}_k(s_t)/\tau) = \mathcal{F}(s_t). \quad (10)$$

Thus $\mathcal{F}(s_t)$ is fully equivalent to the variational problem, with optimal responsibility allocation q^* .

This result further reveals the analytic properties of energy-guided routing. For any two experts i and j ,

$$\log \frac{q_i^*}{q_j^*} = \log \frac{\pi_\phi(i | s_t)}{\pi_\phi(j | s_t)} - \frac{\bar{E}_i(s_t) - \bar{E}_j(s_t)}{\tau}. \quad (11)$$

Therefore, expert responsibility is jointly determined by the routing prior and the expert’s interpretive energy for the current state, with relative differences expressed linearly in the log-responsibility ratio. $\square$

DETAILED IMPLEMENTATION HYPERPARAMETERS

L2DiffM uses a 9-layer EGNN as the hybrid diffusion backbone with a 32-nearest-neighbor graph, $T = 1000$ diffusion steps. The language teacher, MoLlama (a Llama2-based molecular LLM pretrained on ZINC), is fine-tuned on ligand SELFIES via LoRA (rank 8). BCAS uses $\tau_{\text{low}} = 0.36$, $\tau_{\text{high}} = 0.46$, $s_{\text{mid}} = 3$, and $s_{\text{coarse}} = 5$; VSRD-MoE uses $K = 4$ with a linear routing-temperature annealing from $\tau_{\text{init}} = 1.05$ to $\tau_{\text{min}} = 0.12$.

SEMANTIC PRIOR RING-SIZE ANALYSIS

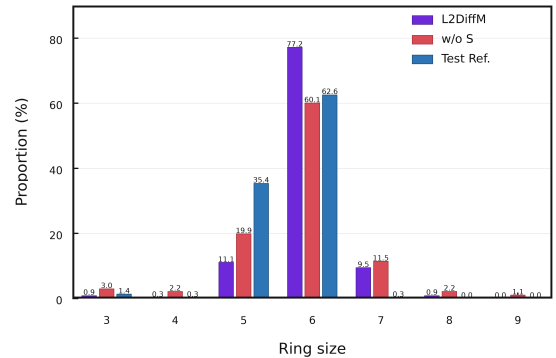


Fig. 1. Ring-size distributions of L2DiffM, w/o S, and reference ligands from the test set.

Fig. 1 compares the ring-size distributions of L2DiffM, w/o S, and the reference ligands from the test set. Introducing the semantic prior markedly reduces non-mainstream small- and large-ring structures and makes the overall distribution closer to that of the reference ligands.

VSRD-MoE EXPERT LOAD ANALYSIS

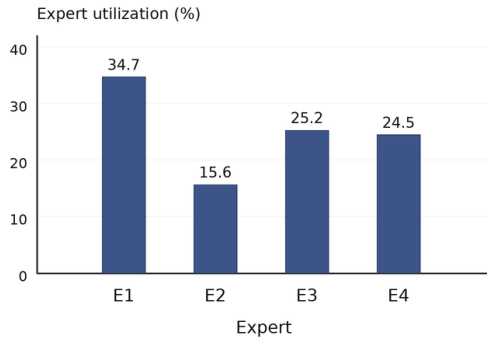


Fig. 2. Average VSRD-MoE expert load on the validation set.

Fig. 2 reports the average validation-set expert load of VSRD-MoE. The non-uniform load distribution indicates that the router assigns differentiated responsibilities to different experts, whereas the non-negligible utilization of all experts shows that the routing process preserves sufficient expert diversity without collapsing to a single dominant path.